Automated Business Analytics Case Study

Rows: 541907 | Columns: 19

Dataset Profile

Column	Dtype	Missing%	Unique	Preview
stock_code	object	0.0	4070	85123A(2313), 22423(2203), 85099B(2159), 47566(1727), 20725(1639)
description	object	0.268	4223	WHITE HANGING HEART T-LIGHT HOLDER(2369), REGENCY CAKEST
quantity	int64	0.0	722	mean=9.552281110965534, min=-80995.0, max=80995.0
invoice_date	datetime64[ns]	0.0	23258	min=2010-12-01 08:26:00, max=2011-12-09 12:50:00
unit_price	float64	0.0	1629	mean=4.6519570590525685, min=0.0, max=38970.0
customer_id	string	0.0	4373	UNKNOWN(135078), 17841.0(7983), 14911.0(5903), 14096.0(5128), 1274
country	object	0.0	38	United Kingdom(495476), Germany(9495), France(8557), EIRE(8196), Sp
invoice_no	object	0.0	25898	573585(1114), 581219(749), 581492(731), 580729(721), 558475(705)
is_cancellation	bool	0.0	2	False(532619), True(9288)
is_return	bool	0.0	2	False(531283), True(10624)
line_total	float64	0.0	6203	mean=18.02868767888217, min=-168469.6, max=168469.6
abs_line_total	float64	0.0	5248	mean=21.338526784116095, min=0.0, max=168469.6
net_revenue	float64	0.0	6203	mean=18.02868767888217, min=-168469.6, max=168469.6
invoice_year	int32	0.0	2	mean=2010.9216083202468, min=2010.0, max=2011.0
invoice_month	int32	0.0	12	mean=7.553126274434543, min=1.0, max=12.0
invoice_ym	datetime64[ns]	0.0	13	min=2010-12-01 00:00:00, max=2011-12-01 00:00:00
has_customer	boolean	0.0	2	True(406829), False(135078)
first_purchase_date	datetime64[ns]	24.926	4270	min=2010-12-01 08:26:00, max=2011-12-09 12:16:00
days_since_first_purchase	float64	0.0	16756	mean=97.90369985455686, min=0.0, max=373.0965277777778

Analysis 1

Question/Goal: Which countries generate the highest net revenue, and how concentrated is revenue among the top five markets?

- United Kingdom leads on net revenue at 8.21M.
- The gap from first to second place is 7.93M versus Netherlands.
- The top five entries account for 96.6% of the shown total, indicating concentration.
- The weakest shown entry is Sweden at 36,596.

country	net_revenue_sum
United Kingdom	8209930.484
Netherlands	284661.54
EIRE	263276.82
Germany	221698.21
France	197403.9
Australia	137077.27
Switzerland	56385.35
Spain	54774.58
Belgium	40910.96
Sweden	36595.91

Analysis 2

Question/Goal: How do monthly net revenue and order count move over time, and where are the major peaks or slowdowns?

- Net Revenue peaks in 2011-11-01 00:00:00 at 1.46M.
- The lowest point is 2011-12-01 00:00:00 at 433,668.
- From 2010-12-01 00:00:00 to 2011-12-01 00:00:00, the metric moves down by 315,289 (-42.1%).
- 2011-11-01 00:00:00 is highest on order count at 3,462.

invoice_ym	invoice_no_nunique	net_revenue_sum
2010-12-01	2025	748957.02
2011-01-01	1476	560000.26
2011-02-01	1393	498062.65
2011-03-01	1983	683267.08
2011-04-01	1744	493207.121
2011-05-01	2162	723333.51
2011-06-01	2012	691123.12
2011-07-01	1927	681300.111
2011-08-01	1735	704804.63
2011-09-01	2327	1019687.622
2011-10-01	2637	1070704.67
2011-11-01	3462	1461756.25
2011-12-01	1015	433668.01

Analysis 3

Question/Goal: Which customers drive the highest lifetime value, and how far ahead are they from the next tier?

- 14646.0 leads on net revenue at 279,489.
- The gap from first to second place is 23,051 versus 18102.0.
- The top five entries account for 66.0% of the shown total, indicating concentration.
- The weakest shown entry is 14096.0 at 57,121.

customer_id	net_revenue_sum
14646.0	279489.02
18102.0	256438.49
17450.0	187482.17
14911.0	132572.62
12415.0	123725.45
14156.0	113384.14
17511.0	88125.38
16684.0	65892.08
13694.0	62653.1
15311.0	59419.340000000004
13089.0	57385.88
14096.0	57120.909999999996

Analysis 4

Question/Goal: What share of activity is tied to returns or cancellations, and which markets look materially worse than baseline?

- USA leads on return rate at 38.5%.
- The gap from first to second place is 21.8% versus Czech Republic.
- The top five entries account for 73.5% of the shown total, indicating concentration.
- The weakest shown entry is Singapore at 3.1%.
- USA is highest on cancellation rate at 38.5%.

country	is_return_mean	is_cancellation_mean	invoice_no_nunique
USA	0.3848797250859107	0.3848797250859107	7
Czech Republic	0.16666666666666666	0.16666666666666666	5
Malta	0.11811023622047244	0.11811023622047244	10
Japan	0.10335195530726257	0.10335195530726257	28
Saudi Arabia	0.1	0.1	2
Australia	0.05877680698967434	0.05877680698967434	69
Italy	0.05603985056039851	0.05603985056039851	55
Bahrain	0.05263157894736842	0.05263157894736842	4
Germany	0.04770932069510268	0.04770932069510268	603
EIRE	0.03684724255734505	0.03684724255734505	360
Poland	0.03225806451612903	0.03225806451612903	24
Singapore	0.03056768558951965	0.03056768558951965	10

Analysis 5

Question/Goal: Which products dominate revenue, and which items appear high-volume but lower-value?

- DOTCOM POSTAGE leads on net revenue at 206,245.
- The gap from first to second place is 41,483 versus REGENCY CAKESTAND 3 TIER.
- The top five entries account for 62.7% of the shown total, indicating concentration.
- The weakest shown entry is JUMBO BAG PINK POLKADOT at 41,620.
- JUMBO BAG RED RETROSPOT is highest on quantity at 47,363.

description	net_revenue_sum	quantity_sum
DOTCOM POSTAGE	206245.48	707
REGENCY CAKESTAND 3 TIER	164762.19	13033
WHITE HANGING HEART T-LIGHT HOLDER	99668.47	35317
PARTY BUNTING	98302.98	18022
JUMBO BAG RED RETROSPOT	92356.03	47363
RABBIT NIGHT LIGHT	66756.59	30680
POSTAGE	66230.64	3003
PAPER CHAIN KIT 50'S CHRISTMAS	63791.94	18902
ASSORTED COLOUR BIRD ORNAMENT	58959.729999999996	36381
CHILLI LIGHTS	53768.060000000005	10229
SPOTTY BUNTING	42065.32	8388
JUMBO BAG PINK POLKADOT	41619.66	21009

Analysis 6

Question/Goal: Where does the business show concentration risk across countries, customers, and products?

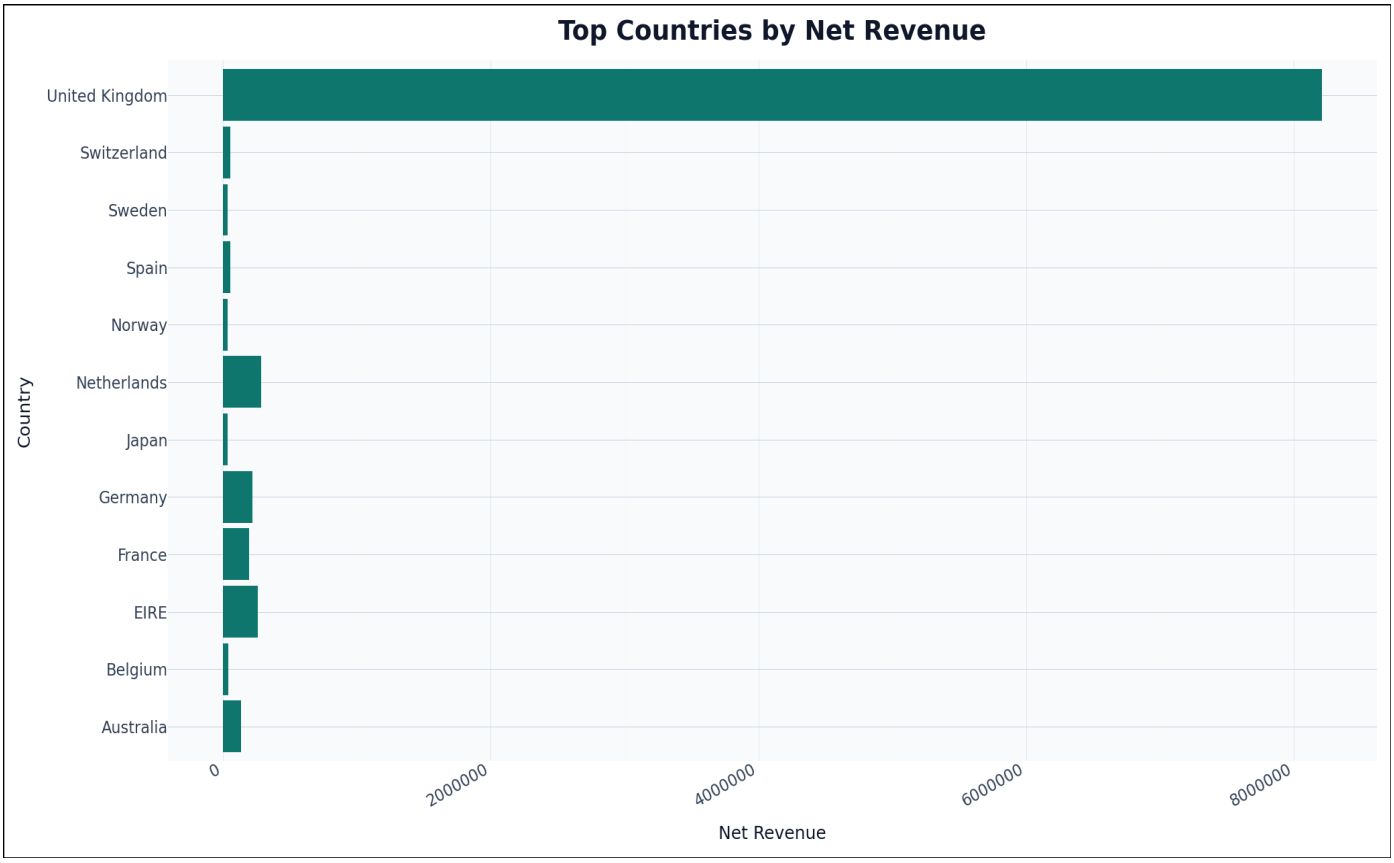
- United Kingdom leads on net revenue at 8.21M.
- The gap from first to second place is 7.93M versus Netherlands.
- The top five entries account for 95.9% of the shown total, indicating concentration.
- The weakest shown entry is Norway at 35,163.
- United Kingdom is highest on order count at 23,492.

country	net_revenue_sum	invoice_no_nunique	customer_id_nunique
United Kingdom	8209930.484	23492	3951
Netherlands	284661.54	101	9
EIRE	263276.82	360	4
Germany	221698.21	603	95
France	197403.9	461	88
Australia	137077.27	69	9
Switzerland	56385.35	74	22
Spain	54774.58	105	31
Belgium	40910.96	119	25
Sweden	36595.91	46	8
Japan	35340.62	28	8
Norway	35163.46	40	10

Visualization 1

Question/Goal: Bar chart of top 12 countries by net revenue with emphasis on concentration

country	net_revenue
United Kingdom	8209930.484
Netherlands	284661.54
EIRE	263276.82
Germany	221698.21
France	197403.9
Australia	137077.27
Switzerland	56385.35
Spain	54774.58
Belgium	40910.96
Sweden	36595.91
Japan	35340.62
Norway	35163.46



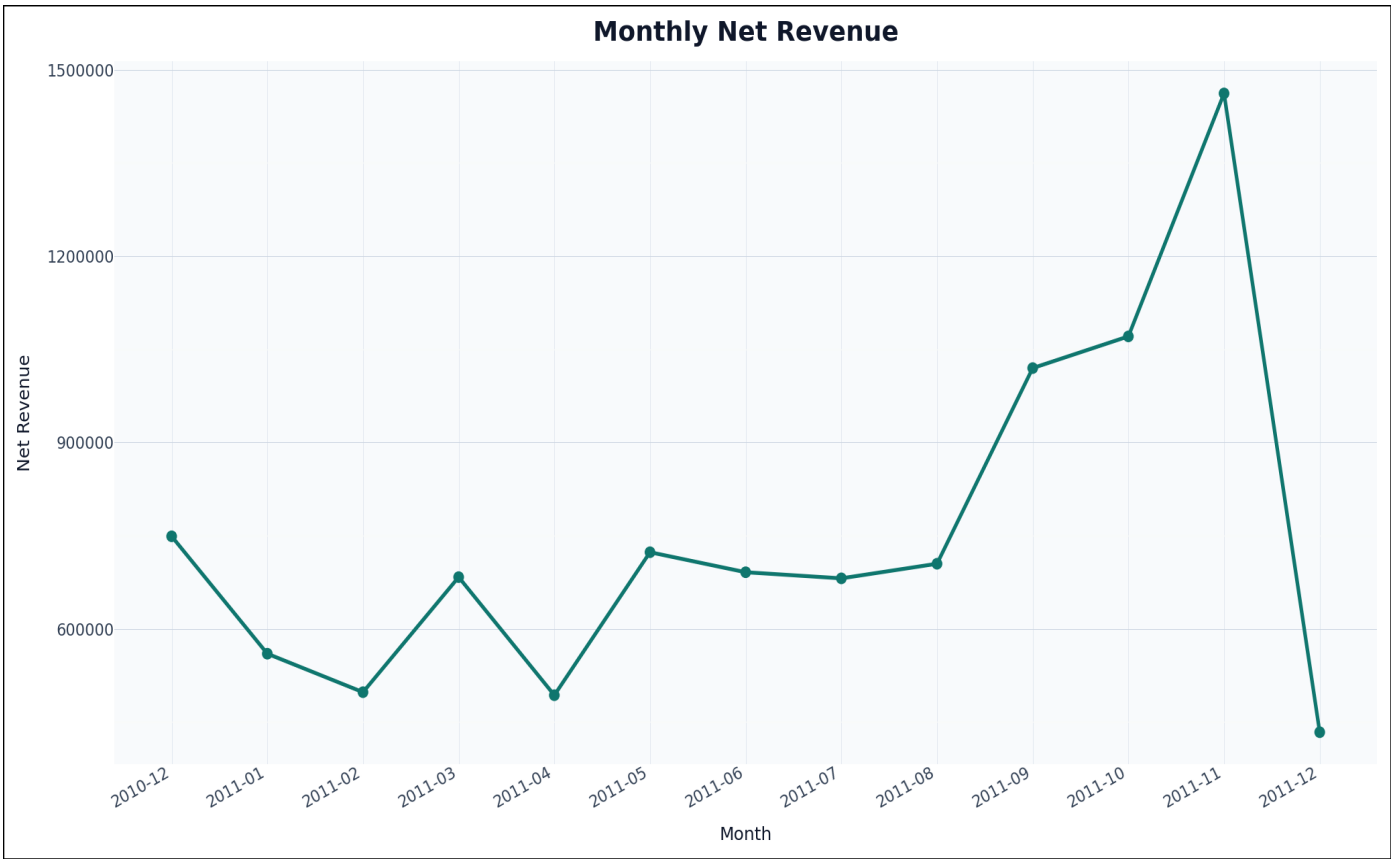
- United Kingdom leads on net revenue at 8.21M.

- The gap from first to second place is 7.93M versus Netherlands.
- The top five entries account for 95.9% of the shown total, indicating concentration.
- The weakest shown entry is Norway at 35,163.

Visualization 2

Question/Goal: Line chart of monthly net revenue over time

invoice_ym	net_revenue
2010-12	748957.02
2011-01	560000.26
2011-02	498062.65
2011-03	683267.08
2011-04	493207.121
2011-05	723333.51
2011-06	691123.12
2011-07	681300.111
2011-08	704804.63
2011-09	1019687.622
2011-10	1070704.67
2011-11	1461756.25
2011-12	433668.01

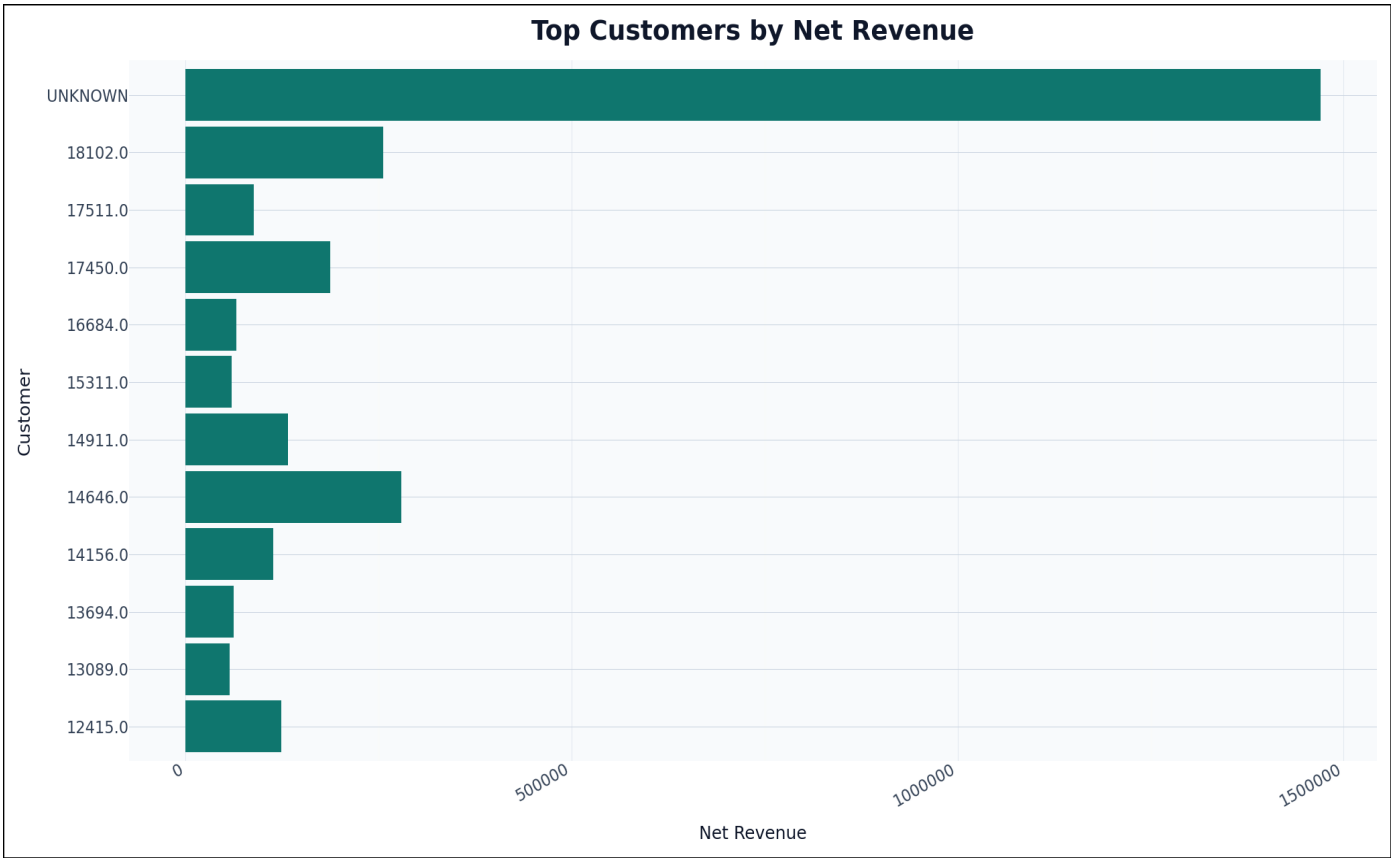


- Net Revenue peaks in 2011-11 at 1.46M.
- The lowest point is 2011-12 at 433,668.
- From 2010-12 to 2011-12, the metric moves down by 315,289 (-42.1%).
- The highest net revenue peak occurred in November 2011 at $1,461,756.25.

Visualization 3

Question/Goal: Bar chart of top 12 customers by net revenue

customer_id	net_revenue
UNKNOWN	1469806.24
14646.0	279489.02
18102.0	256438.49
17450.0	187482.17
14911.0	132572.62
12415.0	123725.45
14156.0	113384.14
17511.0	88125.38
16684.0	65892.08
13694.0	62653.1
15311.0	59419.340000000004
13089.0	57385.88



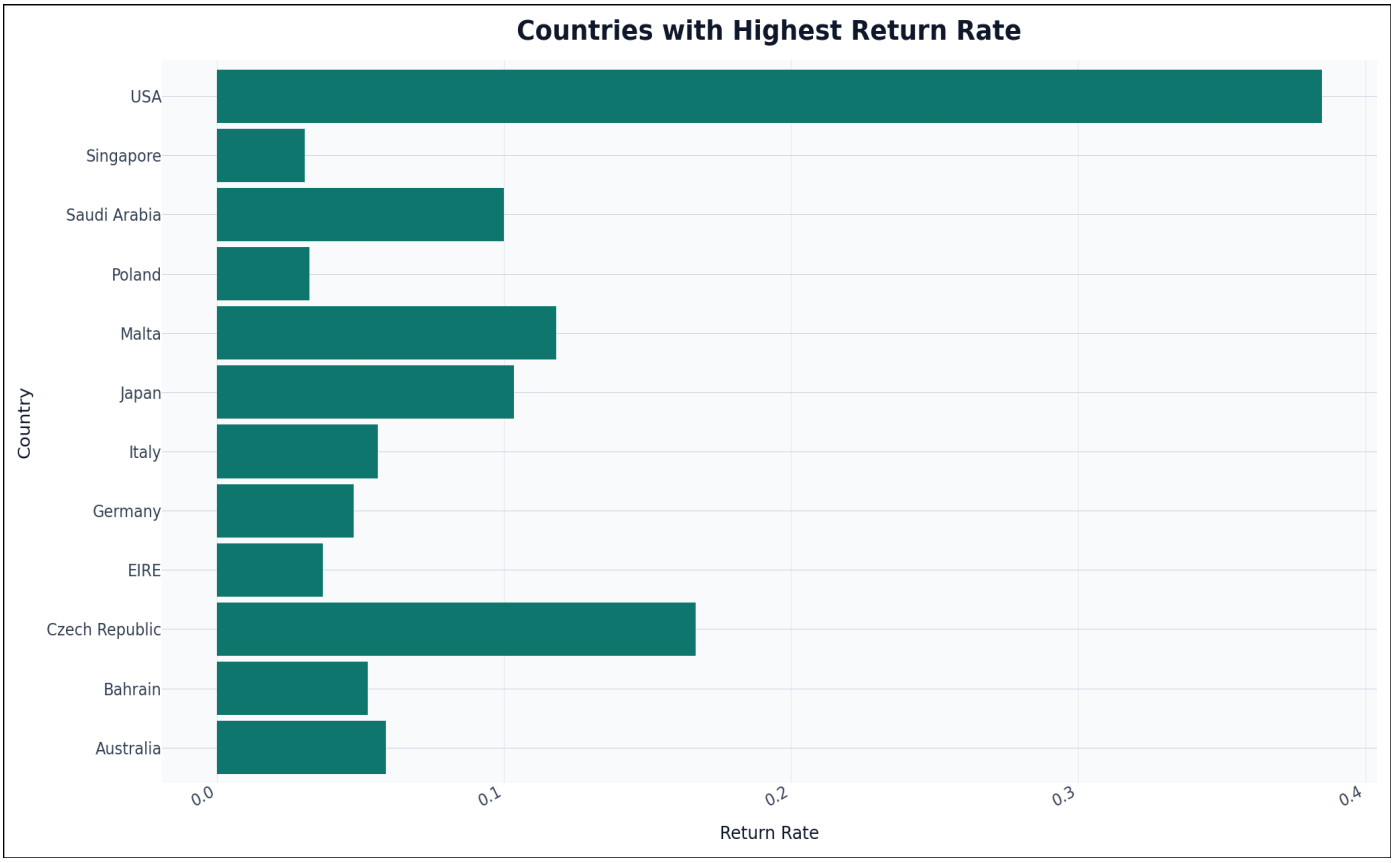
- UNKNOWN leads on net revenue at 1.47M.

- The gap from first to second place is 1.19M versus 14646.0.
- The top five entries account for 80.3% of the shown total, indicating concentration.
- The weakest shown entry is 13089.0 at 57,386.

Visualization 4

Question/Goal: Bar chart of countries with the highest return rate

country	is_return
USA	0.3848797250859107
Czech Republic	0.16666666666666666
Malta	0.11811023622047244
Japan	0.10335195530726257
Saudi Arabia	0.1
Australia	0.05877680698967434
Italy	0.05603985056039851
Bahrain	0.05263157894736842
Germany	0.04770932069510268
EIRE	0.03684724255734505
Poland	0.03225806451612903
Singapore	0.03056768558951965



- USA leads on return rate at 38.5%.

- The gap from first to second place is 21.8% versus Czech Republic.
- The top five entries account for 73.5% of the shown total, indicating concentration.
- The weakest shown entry is Singapore at 3.1%.

Visualization 5

Question/Goal: Bar chart of top 12 products by net revenue

description	net_revenue
DOTCOM POSTAGE	206245.48
REGENCY CAKESTAND 3 TIER	164762.19
WHITE HANGING HEART T-LIGHT HOLDER	99668.47
PARTY BUNTING	98302.98
JUMBO BAG RED RETROSPOT	92356.03
RABBIT NIGHT LIGHT	66756.59
POSTAGE	66230.64
PAPER CHAIN KIT 50'S CHRISTMAS	63791.94
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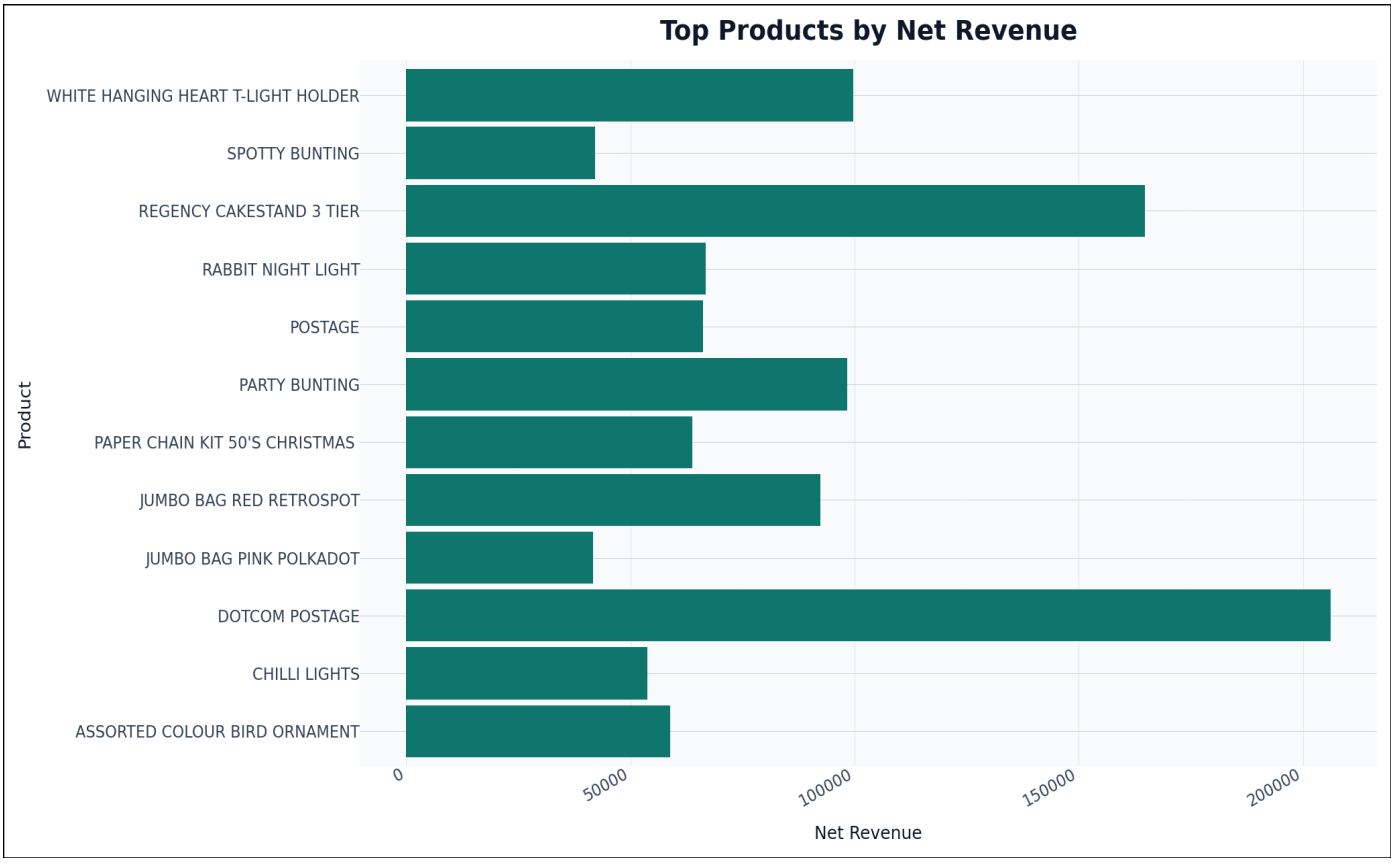
SPOTTY BUNTING	42065.32
JUMBO BAG PINK POLKADOT	41619.66



- DOTCOM POSTAGE leads on net revenue at 206,245.

- The gap from first to second place is 41,483 versus REGENCY CAKESTAND 3 TIER.
- The top five entries account for 62.7% of the shown total, indicating concentration.
- The weakest shown entry is JUMBO BAG PINK POLKADOT at 41,620.